DEGRACE companions in MG-GLAM: status

What landed, what the products are, how they were verified, what the pilots say, what is left

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17 October 2026 merged into `cz`; plan in `docs/`

One week from plan to merged code

- **Scope** (plan of 16 Oct): 42 exact outputs per box (17 primary redshifts + 25 stations at $\Delta\chi = 256\,h^{-1}$Mpc), lens planes instead of snapshots, and the extra products of the brief: δ_L, low-k density and momentum modes, a hashed subsample, f_R at halo centres.
- **Implemented** on `degrace-track-a` (17 commits, 53 files, +10.8k lines): items I1–I10; I11 (optional) not done.
- **Verified**: 180-check regression suite + four unit/integration scripts, six independent code reviews (0 blockers, 8 majors, all fixed), two pilots on ECOSMOG node 2.
- **Merged** into `cz` today (fast-forward, suite green on `cz`).

Take-away

Everything a companion run needs exists, is tested, is on the mainline and has run twice at 1024^3. Three design decisions and one $0.5\,\%$ question remain.

Status of the plan items

item	what	status	evidence
I1	exact-z scheduler accepts arbitrarily close output pairs	**done** (`cz` ae4fa9d)	suite case 7
I2	lens planes: 12 per output, $\Sigma/\bar{\Sigma}$ on 2048^2	**done**	L1–L3
I3–I5	input generator, `OutputSchedule.dat`, per-output classes	**done**	cases 9, 4e; S2 (42 epochs = ECOSMOG to 10^{-7})
I6	provenance per box	partial	JSON lacks generator/LPT/seed (minor)
I7	$\delta_L(k)$ dump, route 1: in our FML IC generator	**done**	V3, V7, V8; `run_deltaL_test.sh`
I8	low-k δ_m and $(1+\delta)$v modes	**done**	V1, V2, mode-by-mode
I9	ID-hashed quantised subsample, 1 in M	**done**	V5, V6; `run_ic2pm_test.sh`
I10	f_R at BDM halo centres	**done**	unit test + f(R) smoke case
I11	optional extras	**open**	not started
pilot	node 2 ΛCDM, $N_g = 2048$	**done**	verified, 0 FAIL
pilot	node 2 f(R), extras + subsample + f_R	**done**	verified, 0 FAIL; 6 h 08 min

All flags default off: without the new `Init.dat` lines the code follows the `cz` path (review 2), apart from `OutputSchedule.dat` and an appended `Setup.dat` block.

Schedule and output classes (I1, I3–I5)

- **17 primary redshifts** $z = 0, 0.125, \ldots, 2$ ($\Delta z = 0.125$), common to all cosmologies.
- **25 stations** at comoving distance $\chi = 256\,k\,h^{-1}$Mpc, $k = 1 \ldots 25$, from the run's own flat matter + Λ $\chi(a)$ (no radiation, $c/H_0 = 2997.92458\,h^{-1}$Mpc); `gen_inputs.py` reproduces every ECOSMOG list to 10^{-7}.
- `#outputs = -42`: every epoch is hit exactly (a step moved onto it or inserted); pairs closer than 0.5 % are allowed. Pilot: 181 steps = 163 ladder + 18 inserted, $\mathrm{d}a_0 = 6 \times 10^{-4}$ from $z = 49$.
- `OutputSchedule.dat`: one line per output: step (= snapshot number), a, z, χ, exact, class.
- **Class 1** (full): $P(k)$, planes, BDM catalogue, f_R at haloes, δ_m and momentum cubes, subsample; **class 2** (station): $P(k)$ and planes. Pilots 17 + 25; production: all 42 class 1.

step	a	z	χ	exact	class
57	0.10275	8.732	6400.0	1	2
63	0.11832	7.452	6144.0	1	2
105	0.33333	2.000	3625.9	1	1
106	0.33796	1.959	3584.0	1	2
127	0.50000	1.000	2312.7	1	1
128	0.50128	0.995	2304.0	1	2
152	0.72727	0.375	1024.9	1	1
153	0.72747	0.375	1024.0	1	2
181	1.00000	0.000	0.0	1	1

Node 2 ($\Omega_m = 0.3$), χ in h^{-1}Mpc. Epochs equal the pinned ECOSMOG list to 10^{-7}; steps 105/106, 127/128, 152/153 are the close pairs.

What a box writes: files per output

file (<step> = snapshot, box 0001)	content	1024^3, $N_g = 2048$ (pilots)	2048^3, $N_g = 4096$
`PowerDM.R/log.<step>.0001.dat`	$P(k)$ on the N_g mesh, linear / log bins, raw CIC	41 KB	80 KB
`LensPlanes.<step>.0001.bin/.meta`	12 planes of $\Sigma/\bar{\Sigma}$, $N^2_{\rm lens}$ float32	201 MB (2048^2)	805 MB (4096^2)
`CatshortV.<step>.0001.DAT` / `.bin` + `.meta`	BDM haloes (in `CATALOGS/`): text, 24 columns; binary twin, 96 B per halo (`CatalogueFormat`)	0.1–0.3 GB (binary $\frac{1}{3}$)	0.3–0.4 GB (binary $\frac{1}{3}$)
`deltam.<step>.0001.bin/.meta`	$\tilde{\delta}_m(\mathrm{k})$ cube, $k_c = 1.5$	469 MB	469 MB
`mom_x,y,z.<step>.0001.bin/.meta`	$[(1+\delta)v_i]^\sim(\mathrm{k})$ cubes, $k_c = 0.75$	3×59 MB (3×138 at $k_c = 1$)	same
`Subsample.<step>.0001.bin/.meta`	1 in $M = 100$ particles, int16	129 MB	1.03 GB
`fR_halo.<step>.0001.bin/.meta`	$(f_R^{\rm centre}, f_R^{\rm mean})$ per halo	9 MB (1.1 M haloes)	∼30 MB
`<IC prefix>_deltaL.bin/.meta`	$\tilde{\delta}_L(\mathrm{k})$ cube, once per box, from the IC job	469 MB	469 MB
`OutputSchedule.dat`, `timing.log`	schedule; per-step timing (with `Run.log`)	KB	KB

- No particle snapshots (`iSave = 0`). Per box: 9.1 GB (ΛCDM pilot, no extras), 25 GB (f(R) pilot, extras at the 17 primaries); with the extras at all 42 outputs ≈ 40 GB at 1024^3 and ≈ 130 GB at 2048^3 (13 TB for 100 boxes).

Conventions shared by every product

- **Epochs.** Positions at the output a ($a_x = a$); velocities half a step behind, $a_v = a - \Delta a/2$ with Δa the step just taken (leapfrog). Every sidecar records `step, a, z, a_x, a_v`. The legacy BDM catalogue converts its velocities with $1/a$ instead of $1/a_v$ (0.3–0.8 % low; decision pending).
- **Units.** Comoving h^{-1}Mpc, peculiar velocities in km/s, masses in $h^{-1}M_\odot$; axes follow the particle x, y, z of the catalogue (`axis_map`).
- **Fourier.** $\tilde{\delta}(\mathrm{k}) = N^{-3}\sum_{\mathrm{x}} \delta(\mathrm{x})\, e^{-i\mathrm{k}\cdot\mathrm{x}}$, $\mathrm{k} = (2\pi/L)(i_x, i_y, i_z)$, $P(k) = L^3\langle|\tilde{\delta}|^2\rangle$; grid node i at $(i-1)L/N$.
- **Mode cube.** $|i_x|, |i_y| \le n$, $0 \le i_z \le n$ with $n = \lfloor k_c L/2\pi \rfloor$; the Hermitian-redundant $i_z = 0$ modes are kept, $\mathrm{k} = 0$ is zero; $(2n+1)^2(n+1)$ complex64 values.
- **Layout.** Fortran `C(0:n,-n:n,-n:n)`, i_z fastest; numpy `fromfile(f, '<c8').reshape(2n+1, 2n+1, n+1)[ix+n, iy+n, iz]`.
- **Mass assignment.** CIC on N_{dep} nodes, never deconvolved inside GLAM; `assignment`, `N_dep` and `interlaced = false` are recorded and offline code divides by W_{CIC}.
- **Sidecar keys.** `product, format, layout, fourier_norm, modes, n, k_c, N_dep, N_grid, N_row, N_p, Box, units, D_code(a)` (the code's flat matter + Λ growth, $D(1) = 1$), `Omega_m, Omega_L, hubble, MG_flag, MG_model, fR0`, and `header` (the PM header: `ic2pm` velocity epoch, lattice tag `IDx/IDz`).

Lens planes (l2): definition, cost, checks

- **Definition.** For projection axis X and slab s of depth $L/N_{\rm slab} = 256\,h^{-1}$Mpc, pixel (i, j) is the mean of $1 + \delta$ over the slab column and the pixel block, i.e. $\Sigma_s(i,j)/\bar{\Sigma}$ with $\bar{\Sigma} = \bar{\rho}_m \times$ depth; δ from the CIC deposit on N_g, not deconvolved. Each plane averages to 1 by construction.
- **Layout.** 12 planes in the order $X_1 \ldots X_4, Y_1 \ldots Y_4, Z_1 \ldots Z_4$; X-planes indexed (y, z), $Y\,(x, z)$, $Z\,(x, y)$; pixel (i, j) covers $[(i-1), i)\,L/N_{\rm lens}$; float32, column-major; block average when $N_{\rm lens} < N_g$.
- **Offline use.** $\kappa = \frac{3}{2}\Omega_m (H_0/c)^2 \sum_{\rm planes} \frac{\chi(\chi_s - \chi)}{\chi_s\, a}\,(\Sigma/\bar{\Sigma} - 1)\,\Delta\chi$, $\Delta\chi = 256\,h^{-1}$Mpc.
- **Cost.** 1.8 s per output, the 201 MB write 12 s: $< 1\,\%$ of a run.

check	result
L1 mass conservation per axis	10^{-7}
L2 vs `ProjectDENSIT`	10^{-5}
L2 36^2 = block mean of 72^2	10^{-5}
L3 P_{2D} vs projected P_{3D}	3σ per ring; 3–7 %
pilots: plane means	1 ± 10^{-7}
pilots: $\Sigma/\bar{\Sigma}$ range at $z = 0$	0.10–51

Open question for Baojiu

Plane normalisation ($\Sigma/\bar{\Sigma}$ vs Σ); do the stations also need halo catalogues?

Fourier dumps, subsample, f_R at haloes (l8–l10)

product	what	verified
`deltam`	$\delta_k = N^{-3}\sum_x \delta(x)e^{-ik\cdot x}$, cube $\lvert k_i\rvert \le k_c = 1.5$, CIC on N_g, complex64 $C(0{:}n, -n{:}n, -n{:}n)$	V1 layout; V2 = `PowerDM.R` to 10^{-4}
`mom_x,y,z`	$(1+\delta)\,v$ on $N_{\rm dep} = N_g/2$, v at $a_v = a - \mathrm{d}a/2$, $k_c = 0.75$ (below $k_{N,\rm dep}/4$: no interlacing)	V6; mode by mode vs decoded particles 7×10^{-5}
`Subsample`	keep 1 in M by splitmix64(ID $\oplus$ salt), int16 positions and velocities, deterministic across twins, threads, restarts	V5 (10^7 IDs), V6
`fR_halo`	$f_R = f_{R,\rm bg}(a_{\rm old})\,u^{n+1}$ at every BDM centre	unit test; f(R) smoke case

- Storage at 1024^3 per output: δ_m 0.47 GB, momentum 0.18 GB ($k_c = 0.75$; 0.41 at 1.0), subsample ($M = 100$) 0.13 GB.
- From the review: `ic2pm` places every particle at array index = Gadget ID and tags the lattice order (`IDz` 2LPTic, `IDx` FML) in the PM header, so the subsample index gives the Lagrangian position.

Definitions: density, momentum and δ_L cubes (I7, I8)

- `deltam`: $\delta(\mathrm{x}) = n(\mathrm{x})/\bar{n} - 1$ from the CIC deposit of the output positions on $N_{\rm dep} = N_g$, forward FFT, cube $|k_i| \le k_c = 1.5\,h\,\mathrm{Mpc}^{-1}$ ($n = 244$ at $L = 1024$: 0.47 GB). Offline $P_{mm}(k) = L^3\langle|\tilde{\delta}_m/W_{\rm CIC}|^2\rangle$ equals `PowerDM.R` bin by bin to 10^{-4} (V2).
- `mom_x,y,z`: $p_i(\mathrm{x}) = \bar{n}_{\rm cell}^{-1}\sum_p v_{i,p}\,W_{\rm CIC}(\mathrm{x} - \mathrm{x}_p) = (1+\delta)\,v_i$, v the peculiar velocity in km/s at a_v; deposit on $N_{\rm dep} = N_g/2$, $k_c = 0.75$ by the rule $k_c \le k_{\rm N,dep}/4 = \pi N_{\rm dep}/4L$ (no interlacing; the f(R) pilot used 1.0). At linear k, $\tilde{v}_i \simeq \tilde{p}_i$.
- `deltaL`: the IC generator's Gaussian field at $z_{\rm ini}$ divided by $D(a_{\rm ini})$ (flat matter + Λ, $D(1) = 1$), i.e. the linear field at $a = 1$; written before any LPT displacement, same cube and layout as `deltam`, identical for a ΛCDM/f(R) twin pair (one IC file). Sidecar adds `seed, z_ini, D_gen(a_ini), pofk_file, lpt_order, fix_amplitude, lattice` (particles at iL/N_p, ID $= 1 + i_x + N_p(i_y + N_p i_z)$) and the cube rms.
- **Uses.** Halo propagator $\Gamma_h(k|M) = \sum \mathrm{Re}[\tilde{\delta}_h\tilde{\delta}_L^*]/\sum|\tilde{\delta}_L|^2$ without cosmic variance; matter propagator $G_m = P_{mL}/P_L$; growth check $\langle\tilde{\delta}_m/W_{\rm CIC}/\tilde{\delta}_L\rangle = D_{\rm code}(a)$ (0.9997–1.0012 at $k \le 0.15$, V3); halo–momentum cross spectra; environment and tidal fields at $R \ge 3/k_c$.
- **Caveat.** Compare dumps with binned means and cross-correlations, not single-mode ratios (CIC of lattice particles is not linear in the displacement; phase scatter $\sim k\Delta/2$ at early times).

Definitions: particle subsample (I9) and f_R at haloes (I10)

`Subsample`

- ID = 0-based lattice index $i + N(j + Nk)$ (i along x for FML ICs, tag `IDx`; Baojiu's 2LPTic files are k-fastest, `IDz`); `ic2pm` places particle = Gadget ID. Lagrangian position $\mathrm{q} = (i, j, k)\, L/N$.
- Keep iff `splitmix64(ID xor SALT) mod M = 0`, `SALT = 0x5EED00DE6C0A2026`, $M = 100$: the same particles at every output and in both twins; $n_{\rm sel} = 10\,741\,150 = N/100$ to 0.03 %.
- Stored in increasing ID: the k-th particle is the k-th ID passing the rule (IDs implicit).
- int16 quantisation: $q = \lfloor 65536\, x/L \rfloor - 32768$, decode $x = (q + 32768 + 0.5)\, L/65536$ ($\Delta x = 15.6\, h^{-1}$kpc); $s = \mathrm{nint}(v/\Delta v)$, $\Delta v = 0.25$ km/s, clipped at ± 32767 (count in the sidecar, 0 in the pilot); v peculiar at a_v.
- Layout: `qx, qy, qz, vx, vy, vz`, $n_{\rm sel}$ each, contiguous; 12 B per particle.

`fR_halo` (f(R) runs, `MG_model = 1`)

- One float32 pair $(f_R^{\rm centre}, f_R^{\rm mean})$ per catalogue row, in `CatshortV` order.
- $f_R^{\rm centre}$: CIC interpolation of f_R at the BDM centre; $f_R^{\rm mean}$: mean over the mesh cells whose centres lie within $R_{\rm vir}$ (equal to the centre value below half a cell).
- $f_R(\mathrm{x}) = f_{R,\rm bg}(a_{\rm old})\, u(\mathrm{x})^{n+1}$, u the solver field (scalar field over its background value) at the pre-step epoch $a_{\rm old}$; $f_{R,\rm bg}(a) = f_{R0}\,[R_0/R(a)]^{n+1}$, $R(a) = 3(\Omega_m a^{-3} + 4\Omega_\Lambda)$.
- Sidecar: f_{R0}, n, $a_{\rm old}$, $f_{R,\rm bg}(a_{\rm old})$, catalogue name and row count.

δ_L from the IC generator (I7, route 1)

- The FML generator writes `<prefix>_deltaL.bin/.meta`: its Gaussian field at $z_{\rm ini}$ divided by $D(a_{\rm ini})$, same layout and convention as `deltam`; $k_c = 1.5$, 0.47 GB.
- Lua: `write_deltaL`, `deltaL_kc`; on in the production input. Costs 1 s.
- Baojiu's ICs and the legacy pilots are out of scope (decision of 17 Oct).

check	result
Zel'dovich: $\delta_L = -ik\cdot\Psi(k)/D$	every mode to 10^{-3}
2LPT ICs	residual 8×10^{-3} (2nd order)
wrong axis / sign / half cell	off by $10^4\times$
V3: ratio to $D(a)$, $k \le 0.15$	0.9997–1.0012
V3: cross-correlation, $k \le 0.05$	0.99998
V3: coordinate offset per axis	$\le 0.03\,h^{-1}$Mpc
V7: 1/2/4 MPI ranks	identical md5

Lesson

CIC density of particles sitting on grid nodes is not linear in the displacement (phase scatter $\sim k\Delta/2$ per mode at early times): compare dumps with binned means and cross-correlations.

Definitions: BDM halo catalogue and $P(k)$ (legacy products)

- `CATALOGS/CatshortV.<step>.0001.DAT`: text, 8 header lines (PM header with the `ic2pm` epoch and lattice tag, a and Δa, step, $N_{\rm row}$, N_g, cosmology, buffer width, mass of the smallest particle, overdensity limit), then one row per halo with 24 columns: x, y, z [h^{-1}Mpc]; V_x, V_y, V_z [km/s]; $M_{\rm bound}, M_{\rm tot}$ [$h^{-1}M_\odot$]; $R_{\rm vir}$ [h^{-1}kpc]; $V_{\rm rms}, V_{\rm circ}$ [km/s]; row index; $c_{\rm vir}$; $N_{\rm part}$; distinct/sub flag; $X_{\rm off}$; $2K/E_p - 1$; spin λ; RadRMS; b/a, c/a; major-axis direction (x, y, z).
- **Binary twin** (`CatalogueFormat = text|binary|both` in `BDM.config`, compiled default binary since 2026-10-18, `text` or `both` on request; 2026-10-17/18): `CatshortV.<step>.0001.bin`, 96-byte little-endian records holding the same 24 columns (22 float32 + 2 int32), same rows and order, lossless; $3\times$ smaller apparent, $\approx 40\,\%$ on disk (this Lustre compresses text $1.9\times$); sidecar `.meta` with columns, units, rows, a, a_v, Δa, cosmology, m_p, `MassMin`, $\Delta_{\rm vir}$, definitions. Reader: `python/glamio` (`read_catalogue`). `BDM.config` is validated at start-up.
- **Halo definition (BDM).** Centre = density peak refined by centre-of-mass iterations; $R_{\rm vir}$ = the spherical-overdensity radius at the Bryan–Norman $\Delta_{\rm vir}$ relative to the mean (338 at $z = 0$, 203 at $z = 1$ for $\Omega_m = 0.3$) plus a resolution correction of up to 0.75 cells; $M_{\rm bound}$ = bound mass inside the uncorrected radius, $M_{\rm tot}$ = all mass inside $R_{\rm vir}$; `MassMin` $= 10^{11}$ raised to 20 particles ($1.7 \times 10^{12}\,h^{-1}M_\odot$ at 1024^3); rows in peak order, not by mass; velocities converted with $1/a$ (low by a_v/a, decisions frame).
- `PowerDM.R.<step>.0001.dat`: 2 header lines (a, step, z, N_g, $k_{\rm Ny} = \pi N_g/L$), then bin, k [$h\,{\rm Mpc}^{-1}$], number of modes, P [$(h^{-1}{\rm Mpc})^3$] in linear bins of width $2\pi/L$; δ from CIC on N_g, raw: divide by the shell-averaged $W^2_{\rm CIC} \simeq 1 - \frac{2}{3}\sin^2(\pi k/2k_{\rm Ny})$ offline (as in the twin comparison). `PowerDM.log`: the same in logarithmic bins.
- `timing.log`: per-step wall time split into Force, Move, Density, IO, BDM, Analysis, MG (source of the cost figures).

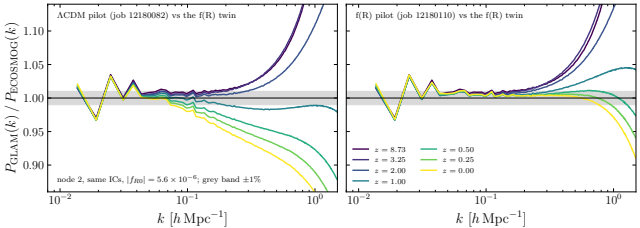
Pilot: node 2 ΛCDM, $N_g = 2048$ (job 12180082)

wall time	2 h 32 min, 64 cores, `cosma8-serial`
CPU efficiency	83 %
peak memory	118 GB (91 GB `PMP2main`)
steps	181, median 0.60 min
outputs	42 $P(k)$, 42 plane sets, 17 catalogues
disk per box	9.1 GB
core-hours	≈ 165 per ΛCDM box

step type	mean wall time
ordinary step	0.60 min
station output: 2048^3 $P(k)$ 25 s, planes 1.8 s, write 12 s	0.96 min
primary output: + BDM halo finder	2.6 min
all 42 outputs	43 min (29 %)
of which lens planes	$< 1\,\%$

- `check_run.py`: schedule, epochs, files, sizes, plane means, catalogues, ic2pm epoch: 0 FAIL.
- BDM dominates the output cost; the new products are cheap. The extras (modes, subsample, f_R) add at most 1.4 min per primary output on the f(R) pilot (two frames on).

Pilots vs the ECOSMOG twin: same ICs, mode-exact



- Left, ΛCDM vs the f(R) twin: flat $+0.5$ to $+0.7\,\%$ at $k < 0.1$ at every epoch, already at $z = 8.7$; late times: ECOSMOG above us by 5 % ($k = 0.1$–0.3) and 10 % ($k = 0.3$–1), the f(R) enhancement for $|f_{R0}| = 5.6 \times 10^{-6}$.
- Right, f(R) vs f(R), the real twin test: the same $+0.5$ to $+0.8\,\%$ at $k < 0.1$; within $\pm 1\,\%$ to $k = 0.3$ at every epoch; $-0.6\,\%$ at $k = 0.3$–1 at $z = 0$ (PM force resolution). $k > 0.3$ at $z > 3$: lattice discreteness.

The $+0.5\,\%$ at linear scales is the same in both gravity models

A code or ingestion offset, not physics: `ic2pm`'s half-step convention, POWMES normalisation or a 0.25 % growth difference; a $\mathrm{d}a_0 = 3 \times 10^{-4}$ restart of 20 steps isolates the half-step term.

f(R) pilot (job 12180110): completed and verified

	f(R), extras on (ΛCDM pilot)
wall, 64 cores	6 h 08 min (2 h 32 min)
CPU efficiency	83 % (83 %)
MaxRSS	177 GB (118 GB)
non-output step	1.72 min; multigrid 1.1 min (0.60 min)
station output	$+0.37$ min: P(k), planes ($+0.36$)
primary output	$+2.9$ min: BDM + extras ($+2.0$, BDM)
42 outputs	59 min = 16 % of the run (29 %)
disk per box	25 GB (9.1 GB)

- `check_run.py`: 42 exact outputs, 42 P(k) + planes, 17 catalogues, 0 FAIL, 0 WARN.
- Per primary: δ_m cube $n = 244$ (469 MB), momentum cubes $n = 162$ (3×138 MB), subsample of 10.74 M particles $= N/100$ to 0.03 %, no clipping (129 MB), f_R at 1.12 M haloes at $z = 1$ (9 MB): 1.0 GB.
- V9: the extras cost at most 1.4 min per primary (the excess over the ΛCDM primaries), ≤ 23 min $= 6\,\%$ of this run and $\leq 15\,\%$ of a ΛCDM run; an upper bound until `ExtraOutputs` gets per-product timers. Memory 177 GB on a 1 TB node.
- ≈ 390 core-hours per f(R) box at 1024^3.
- Caveats: $k_{c,\mathrm{mom}} = 1.0$ (use the momentum modes below $0.78\,h\,\mathrm{Mpc}^{-1}$); the binaries predate the review fixes.

Independent review: six reviewers, eight majors, all fixed

review	B	M	m	n
1 Fortran extras module	0	0	9	5
2 scheduler, parameters, build	0	2	6	6
3 particle subsample	0	1	4	6
4 δ_L generator dump	0	0	7	7
5 Python tools and tests	0	2	10	6
6 docs and provenance	0	3	11	8

Self-contained prompts and findings in `docs/reviews/degrace-track-a/`; rerunnable.

Majors and their fixes

- subsample "ID" was the IC file order $\rightarrow$ `ic2pm` scatters by Gadget ID and tags the lattice order
- optional `Init.dat` lines defaulted silently when malformed $\rightarrow$ stop with the line
- f_R gate on `MG_flag` only $\rightarrow$ also `MG_model = 1`
- twin script skipped early epochs (5-decimal header) $\rightarrow$ epochs from the schedule file
- no test sensitive to axis order/sign of the dumps $\rightarrow$ mode-by-mode test
- plan flags table, catalogue velocity epoch, interlacing status $\rightarrow$ documented, rules added

After the fixes: suite 180/180, four scripts green, merged.

Decisions pending and open items

Decisions (yours)

- **BDM catalogue velocities** are converted with $1/a$, the new products with $1/a_v$: the catalogue is low by a_v/a, $0.8\,\%$ ($z = 2$) to $0.3\,\%$ ($z = 0$). Fix the legacy finder (changes every GLAM catalogue) or keep documenting it (multiply by a/a_v offline; `glamio` does).
- **Momentum deposit** at $N_g = 4096$ f(R): the transient $N^3_{\rm dep}$ buffer (34 GB) does not fit the 985 GB runs; reuse `FI` or restrict extras to $N_g = 2048$.
- **Interlacing**: not implemented; default $k_{c,\rm mom} = 0.75$ stays below $k_{N,\rm dep}/4$ and PMP2init refuses more. Implement if $k > 0.78$ momentum modes are wanted.
- **Disk**: 1.5 TB freed on 17 Oct (validation IC, February checkpoints, BDM PM snapshots); still to decide: the 21 BDM replay density tapes and membership tapes (195 GB on disk), 15 GB of superseded fiducial catalogues.

Open minors (grouped in `TRIAGE.md`): parameter-path robustness, f_R product details, provenance JSON, test strength, docs drift. **Questions for Baojiu**: BDM cadence, plane normalisation, CAMB ω_c pitfall, r2g velocity convention, node 900.

Next steps

1. Explain the $+0.5\,\%$: the $\mathrm{d}a_0 = 3 \times 10^{-4}$ restart test (half-step term) and a POWMES normalisation check; add per-product timers to `ExtraOutputs` (the V9 split).
2. Settle the three decisions above; rebuild the frozen pilot binaries from `cz`.
3. Production recipe per node: `gen_inputs.py --node N --gravity {lcdm,fr}` $\rightarrow$ `submit.sh` (`ic2pm half` on Baojiu's ICs) $\rightarrow$ `check_run.py`; ≈ 165 core-hours per ΛCDM box, ≈ 390 for f(R); 9 GB (ΛCDM) or 25 GB (f(R) with the extras) per box.
4. Readers for the new products in `halocat` (planes, modes, subsample, f_R), following the `.meta` keys.
5. Reply to Baojiu with the cross-check of the 42 epochs and the open questions.

Summary

- The DEGRACE companion outputs (schedule, lens planes, δ_L, modes, subsample, f_R) are implemented, tested, reviewed and merged into `cz`.
- The ΛCDM pilot runs in 2.5 h per box with all 42 outputs, the f(R) pilot with all the extras in 6 h; the new products cost about 1 % of a run, the halo finder 23 %, the extras at most 6 %.
- Both pilots agree with the ECOSMOG twin to $+0.5$–$0.8\,\%$ at linear scales at every epoch, and the f(R) run reproduces the f(R) spectrum to $\pm 1\,\%$ up to $k = 0.3$; the small constant offset is common to both gravity models and is the one question still open.
- Three design decisions stand between here and production.